

CLC+ Core User Manual Annex

CLC+ Core production and provision of
complementary consultancy



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4.0	2023-12-22	CLC+ Core consortium (CLF, GV, GAF)	Update of the entire document with all newly implemented features and improvements of the system (within SC8)

Applicable Documents (AD)

ID	Document Name / Content
AD01	Tender Specifications – EEA/DIS/R0/20/019 – Copernicus Land Monitoring Service – Production of the CLC+ Core and Provision of Complementary Consultancy Services
AD02	Consortium's Technical Proposal – in response to Call for tenders EEA/DIS/R0/20/019 – CLC+CORE Technical Proposal - v2.0
AD03	Specific Contract No 3436/R0-COPERNICUS/EEA.58286

ID	Document Name / Content
AD04	Minutes of the KOM (CLC+Core_KOM_Minutes.docx)
AD05	PM1 - Project Management and Governance Plan (CLC+Core_Deliverable_PM1_Project-Management_Plan_V1.0.pdf)
AD06	Implementation of CLC+ based on the EAGLE concept –Additional support for further development of CLC+databases (CLC+ and CLC+ instances, namely CLC+ LULUCF instance and CLC+ Legacy instance) - Task 4: From CLC+ Core to CLC+ Legacy - 3436/R0-Copernicus/EEA.57755
AD07	PM2 - 1st Project Management and Status Report (CLC+Core_Deliverable_PM2_1st-Project-Management-Status-Report_V1.0.pdf)
AD08	CLC+ Core Decision Log_V1.4 (CLC+Core_Decision_log_V1.4.pdf)
AD09	D1 - Software Design (CLC+Core_D1_Software_Design_V1.0.pdf)
AD10	D3.2.1 - Demonstration and documentation of ingested CLMS datasets, under this Service Contract (CLC+Core_Deliverable_D3.2.1_Demonstration-and-documentation-of-ingested-CLMS-datasets_V01.pdf)
AD11	D3.1 - Documentation of deployment of CLC+ Core products to DIAS – cloud service (CLC+Core_Deliverable_D3.1_Documentation of deployment of CLC+ Core products to DIAS_V1.0.pdf)
AD12	LULUCF_classes_simplified_EAGLE_query_rules_20210325.xlsx (Provided by EAGLE Group)
AD13	Specific contract no 3506_R0-COPERNICA_EEA.59433

1. Admin User only

In this section information is given which is relevant for **Admin Users only**. This mainly concerns the categories Organisations and Users, where the Admin User has additional rights compared to a normal System User. An important task given to the admin user is the possibility to upload a new EAGLE ontology into the system.

1.1 Organisations

This navigation menu tab provides an overview of the registered **Organisations** of the CLC+ Core system.

Organisations are taken over by default from EIONET on your first login to CLC+ Core. Organisations play an important role when you define the visibility of your Ingestion or Extraction for publishing it. The visibility can be limited to users from your organisations only or for the country of the selected Organisation. Further, the Admin User has rights to add or remove Organisations (see [Figure 6-1](#)).

As an **Admin User** your account has certain authorizations, i.e. you are able to add new Organisations.

Search function, to **search for an organisation** within the system.

EIONET ID	Name ↑	E-Mail	URL	Country	Status
-	CLC				ACTIVE
at_cloudflight	Cloudflight		https://www.cloudfligh...	Austria/Österreich	ACTIVE
eu_eea	European Environment ...		http://www.eea.europa...	European Environment /	ACTIVE
de_gaf	GAF AG		https://www.gaf.de/	Germany/Deutschland	ACTIVE

Items per page: 10 1 - 4 of 4 |< < > >|

Figure 6-1: Menu item – Organisations

1.2 Users

This navigation menu tab provides an overview of the registered User of the CLC+ Core system.

Users are taken over by default from EIONET on your first login to CLC+ Core. Depending on your role assigned, you have more or less rights. The Admin User has rights to add or remove Users from the system or change their status. To open your Profile, click on your name (see [Figure 6-2](#)). Now you will be forwarded to the Profile view (section 1.5.1 in User Manual).

As an Admin User your account has certain authorizations, i.e. you are able to add new Users.

Further, you have the possibility to **search for a User**.

EIONET Username	↑ Last name	First name	Organisation/s	Email	Roles	Status
clccore	CLC+ Core	User Admin/S...	CLC, European Environ...	thomas.mathis@cloudf...	EAGLE maintainer / Approver...	ACTIVE
hochpmat	Hochpochler	Matthias	Cloudflight	-	User Administrator / Support	ACTIVE
jaffrdav	Jaffry	David	Cloudflight	-	Database Administrator, EAG...	ACTIVE
kurzmmar	Kurzmann	Markus	Cloudflight	-	-	ACTIVE
lindmame	Lindmayer	Amelie	GAF AG	amelie.lindmayer@gaf...	User Administrator / Support	ACTIVE
mathitho	Mathis	Thomas	Cloudflight	-	-	ACTIVE
rennetho	Renner	Thomas	GAF AG	thomas.renner@gaf.de	-	ACTIVE

Items per page: 10 1 - 7 of 7 |< < > >|

Figure 6-2: Menu item – Users.

1.2.1 User Profile

To open the **User Profile** you can either click on the User Profile Icon in the Header, which opens a context menu with the actions 'My Profile' and 'Logout' ([Figure 6-3](#)) or in the Users tab by clicking on your name ([Figure 6-3](#)). You will then be forwarded to the Profile view.

The screenshot shows the 'Data Catalogue' interface. At the top, there are navigation links: Data Catalogue, EAGLE Ontology, About EAGLE, Organisations, and Users. Below these is a filter bar with buttons for 'Filter', 'My organisation/s', 'My data', 'Ingestions', and 'Extractions'. A search bar is also present. On the right, there are buttons for 'ADD INGESTION' and 'ADD EXTRACTION'. A user profile dropdown menu is open, showing options: 'My Profile', 'Version:', and 'Logout'. Below the filter bar is a table with columns: EAGLE Approved, Name, Type, Created At, Created By, Country, Region, Reference Year, Time Range, Organisation/s, Contact Person, Status, and INSPIRE Themes. The table contains one row of data.

EAGLE Approved	Name	Type	Created At	Created By	Country	Region	Reference Year	Time Range	Organisation/s	Contact Person	Status	INSPIRE Themes
CLC+ Backbone (2018)			05.09.2023	CLC+ Core User Admin...	European Environ...		2018	01.07.2017 - 30.06.2019	CLC, European En...	copernicus@eea...	USED	Land cover

Figure 6-3: User Profile – Actions.

1.2.2 User Profile View

When you click on the context menu action 'My Profile' of the User Profile Icon you will find yourself within the User Profile view. Thus, a profile can only be changed by a person with the role 'User Administration / support'. Within your Profile you can see several information (see [Figure 6-4](#)) such as:

General Information: In this area the most important information from your user profile is shown which is username, first and last name, which organisation you are part of and your current status. The EIONET button guides you to the [Eionet Portal](#).

Roles: In this area it is displayed what roles are assigned to you. Roles define what you as a user are allowed to do within the CLC+ Core system from a functional point of view. Typically, roles are for only Users which administer the application.

As an Admin User can be deactivated by a person with the role 'User Administration / support'. "Inactive" Users do not have access to the application.

The screenshot shows the 'User Profile' view for 'User CLC+ Core User Admin/Support'. At the top, there are navigation links: Data Catalogue, EAGLE Ontology, About EAGLE, Organisations, and Users. Below these is a header bar with the user's name and a status indicator 'ACTIVE'. There are buttons for 'DEACTIVATE' and 'EDIT'. The profile is divided into two main sections: 'General Information' and 'Roles'.

General Information

EIONET Username: clccore

Last name: CLC+ Core

First name: User Admin/Support

Organisation/s: CLC, European Environment Agency

Email: thomas.mathis@cloudflight.io

Status: active

[VISIT EIONET FOR CONTACT DETAILS](#)

Roles

Name	Description
EAGLE maintainer / Approver	As an EAGLE maintainer you are allowed to add a new EAGLE ontology version to the CLC+ Core and approve EAGLE compliance of ingestions.
User Administrator / Support	As an User Administrator / Support you are allowed to manage (Add, Edit, Activate/Inactivate) users and organisations within CLC+ Core. Additionally, you can view and edit any Ingestion and Extraction within the CLC+ Core in order to support users if they are in need of help.

Figure 6-4: View User Profile (User Administrator).

1.2.3 Edit User Profile

Editing your profile will only affect / edit the user profile in the CLC+ Core system and not the EIONET account. **General:** Changing your first- and last name and the organisation can only be done by a person with the role 'User Administration / support'.

Roles can only be changed by a person with the role 'User Administration / support'.

By clicking on 'Save' your entered information will be saved. If you do not want to save the changed information, then you have the possibility to click on 'Cancel' (see [Figure 6-5](#)).

General Information		Roles
EIONET Username clccore		
Last name *	CLC+ Core	EAGLE maintainer / Approver As an EAGLE maintainer you are allowed to add a new EAGLE ontology version to the CLC+ Core and approve EAGLE compliance of ingestions.
First name *	User Admin/Support	User Administrator / Support As an User Administrator / Support you are allowed to manage (Add, Edit, Activate/Inactivate) users and organisations within CLC+ Core. Additionally, you can view and edit any Ingestion and Extraction within the CLC+ Core in order to support users if they are in need of help.
Organisation *	CLC European Environment Agency	
Email *	thomas.mathis@cloudflight.io	

Figure 6-5: Edit User Profile.

1.3 Add / upload new EAGLE Ontology

In the "Add new version" dialog a call is made to the EIONET data dictionary. This feature is for expert user only. The dictionary is immediately compared to the previous version and differences will be updated. For example, as in the figure below, there are no differences, which is why you get this view ([Figure 6-6](#)).

By clicking on the **EIONET Data Dictionary** link you will be redirected to the EIONET data dictionary. Data Dictionary¹ holds definitions of datasets, tables and data elements (see [Figure 6-7](#)). Each of these three levels is defined by a set of attributes, the core set of which corresponds to ISO 11179 standard for describing data elements. The whole attribute set is flexible, and attributes can be added / removed from/to the system.

✓ Added Elements (339) ✓ Updated Elements (154) ✓ Deleted Elements (135)

Land Cover Components (LCC)

- Abiotic, Non-Vegetated Surfaces and Objects (LCC-1)
 - Natural Material Surfaces (LCC-1_2)
 - Unconsolidated Surfaces (LCC-1_2_2)
 - Natural Deposits (LCC-1_2_2_3)
 - Organic Deposits, Peat (LCC-1_2_2_3_2)

Figure 6-6: Add new EAGLE Ontology Version.

¹<https://dd.eionet.europa.eu/documentation/doc1>

European Environment Agency

Login | | | >> Eionet portal

EIONET Data Dictionary

You are here: [Eionet](#) » [Data Dictionary](#) » [Vocabulary](#)

- Help and documentation
- Datasets
- Tables**
- Data elements
- Schemas
- Vocabularies**
- Services
- Namespaces

Vocabulary: *EAGLE Nomenclature*

[← Back to set](#) [▼ Exports](#)

Folder	landcover (Landcover nomenclatures)
Identifier	eagle
Label	EAGLE Nomenclature
Base URI	https://dd.eionet.europa.eu/vocabulary/landcover/eagle/
Registration status	Public draft 04 Nov 2021 15:54:57
Type	Common
Version	3.1.2

Figure 6-7: EIONET data dictionary.

Once a new EAGLE ontology version is published all users get a notification (see section 1.7.3 in User Manual).

1.4 Approval EAGLE barcoding

The EAGLE barcoding of an Ingestion can be approved for being **EAGLE compliant** by any user with the additional role of being an “EAGLE Maintainer/Approver”. When opening an Ingestion, the button “Approve EAGLE Compliance” can only be seen by users with that role (see Figure 6-8). The User gets notified once the EAGLE barcoding for the Ingestion is approved by an EAGLE Maintainer.

Data Catalogue | EAGLE Ontology | About EAGLE | Organisations | Users

CLC+ Core User Admin/Support

Data Catalogue / Ingestion 'UA_IBK_2018'

Ingestion 'UA_IBK_2018' | [INGESTED_PREVIEW](#) | [REPUBLISH ON GEOSERVER](#) | [START INGESTION](#) | [PUBLISH](#)

[APPROVE EAGLE COMPLIANCE](#)

General Information

[EDIT](#)

Name
UA_IBK_2018

Country
Austria/Österreich

Region
Innsbruck

Reference Year
2018

Time Range of Ingested Data
01.01.2018 - 31.12.2018

INSPIRE Themes
[LU Land use](#)
[LC Land cover](#)

Created By
[CLC+ Core User Admin/Support](#)

Organisation/s
[CLC](#)
[European Environment Agency](#)

Contact Person
-

Link to Document/s
-

Description

Approve EAGLE compliance for 'UA_IBK_2018'?

You can approve this ingestion as compliant with EAGLE. Other users can identify verified ingestion by a specific tag which is shown near to the name.

The creator of the ingestion is notified about the approval.

[CANCEL](#) [APPROVE](#)

Input Classes

<https://services.ga-cliccore.cloudflight.host/cliccore/9474ed82-9a79-4950-88d4-...>

Class Code	Name *	EAGLE Elements *	100% EAGLE compliant	Colour	Show in Map
11100	Name * 11100	Sealed Artificial Surfaces and Constructions	+10		
11210	Name * 11210	Sealed Artificial Surfaces and Constructions	+10		
11220	Name * 11220	Sealed Artificial Surfaces and Constructions	+11		
11230	Name * 11230	Sealed Artificial Surfaces and Constructions	+11		

[DOWNLOAD EAGLE BARCODING](#)

Figure 6-8: Approval of EAGLE compliance for an Ingestion.

The Ingestion gets than a quality stamp in the Data Catalogue (first column) and when opening the Ingestion next to the status. Please note that for now it is only planned to go through the approval process for the CLMS products.

The **EAGLE approval** for an Ingestion can also be revoked again by the EAGLE Maintainer (see [Figure 6-9](#)). Each approval change is saved and displayed in the history of an Ingestion.

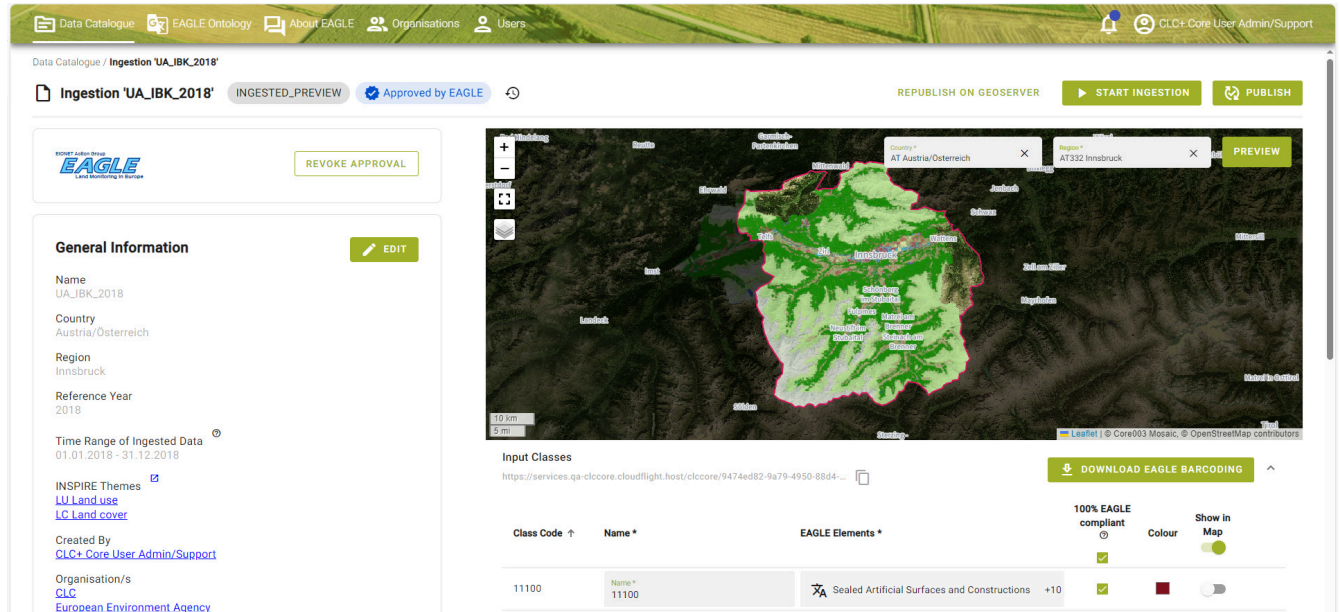


Figure 6-9: Approval and possibility to revoke the EAGLE approval again.

2. List of abbreviations

Abbreviation	Name	Reference
ADs	Applicable Documents	
AI	Action Item	
API	Application Programming Interface	
CLC	CORINE Land Cover	https://land.copernicus.eu/en/products/corine-land-cover
CLC+	CORINE Land Cover +	https://land.copernicus.eu/en/products/clc-a-new-generation-land-information-system-for-europe
CLMS	Copernicus Land Monitoring Service	https://land.copernicus.eu/en
COG	Cloud-optimized GIF	
CORINE	Coordination of information on the environment	
CZ	Coastal Zones	
DB	Database	
DIAS	Copernicus Data and Information Access Services	

Abbreviation	Name	Reference
EAGLE	Eionet Action Group on Land monitoring in Europe	https://land.copernicus.eu/en/eagle
EC	European Commission	https://commission.europa.eu/
EEA	European Environment Agency	https://www.eea.europa.eu
EEA39	The 33 member and 6 cooperating countries of the EEA	
EEA38+UK	The 32 member and 6 cooperating countries of the EEA + United Kingdom	
EO	Earth Observation	
ESA	European Space Agency	https://www.esa.int
ETC	European Topic Centre	https://www.eionet.europa.eu/etcs
EU	European Union	
EU27	The 27 Member States of the European Union	
FM	Final Meeting	
FWC	Framework Contract	
GDAL	Geospatial Data Abstraction Library	
GDB	Geodatabase	
GIO	GMES Initial Operations	
GPKG	GeoPackage	
HDF5	Hierarchical Data Format	
HRL / HRLs	High Resolution Layer / High Resolution Layers	
IaaS	Infrastructure-as-a-Service	
ID	Identification Number	
INSPIRE	Infrastructure for Spatial Information in Europe	https://knowledge-base.inspire.ec.europa.eu/index_en
ISO	International Organisation for Standardization	
ITT	Invitation to Tender	
JDBC	Java Database Connectivity	
JWT	JSON Web Token	
KOM	Kick-Off Meeting	
LC	Land Cover	
LCC	Land Cover Component	
LCH	Land Characteristics	
LC/LU	Land Cover / Land Use	
LCC	Land Cover Component	
LU	Land Use	

Abbreviation	Name	Reference
LUA	Land Use Attribute	
LULUCF	Land Use, Land Use Change and Forestry	
LYR	ArcGIS Layer File	
MMU	Minimum Mapping Unit	
MS	Member States	
NFRs	Non-Functional-Requirements	
NRC	National Reference Centre	
NUTS	Nomenclature of Territorial Units for Statistics	
N2K	Natura 2000	
OBDC	Open Database Connectivity	
PDF	Portable Document Format	
PM	Progress Meeting	
PMP	Project Management Plan	
PoC	Proof of Concept	
QA	Quality Assurance	
QC	Quality Control	
QM	Quality Management	
QML	QGIS Style file	
REST	Representational State Transfer	
RZ	Riparian Zones	
SC	Specific Contract	
SHP	ESRI Shapefile	
SLD	Styled Layer Descriptor	
SQL	Structured Query Language	
UA	Urban Atlas	
UI	User Interface	
URI	Uniform Resource Identifier	
URL	Uniform Resource Locator	
UUID	Universally Unique Identifier	
VM	Virtual Machine	
WBS	Work Breakdown Structure	
WEKEO	Copernicus DIAS reference service for environmental data, virtual environments for data processing	https://www.wekeo.eu/
WP	Work Package	
ZIP	ZIP Format	